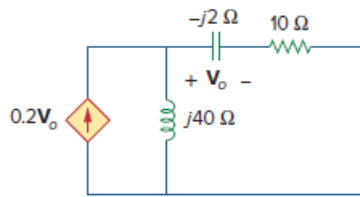


Problem

Calculate the output impedance of the circuit shown in Fig. 10.102.

Figure 10.102:



Step-by-step solution

Step 1 of 5

Draw the following circuit diagram:

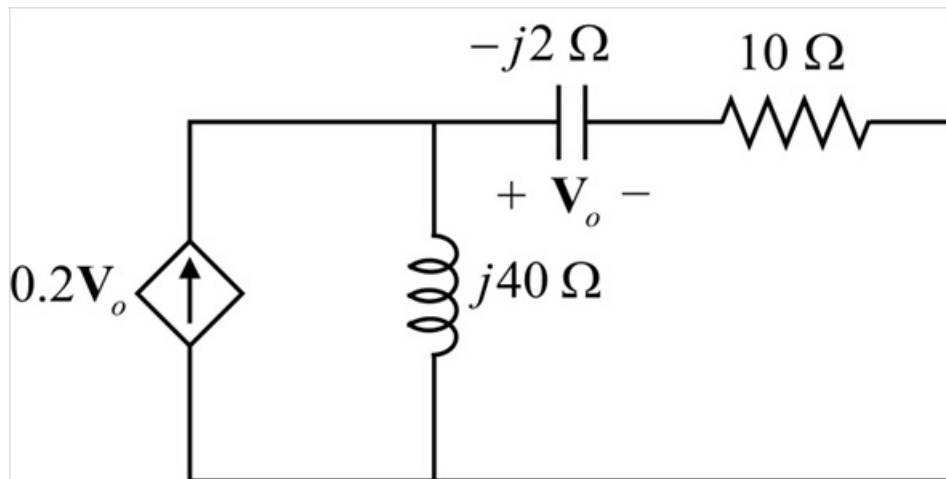


Figure 1

Step 2 of 5

Apply a dummy voltage source $I_s = 1 \text{ A}$ to the circuit in Figure 1 to find the output impedance.

Draw the modified circuit diagram:

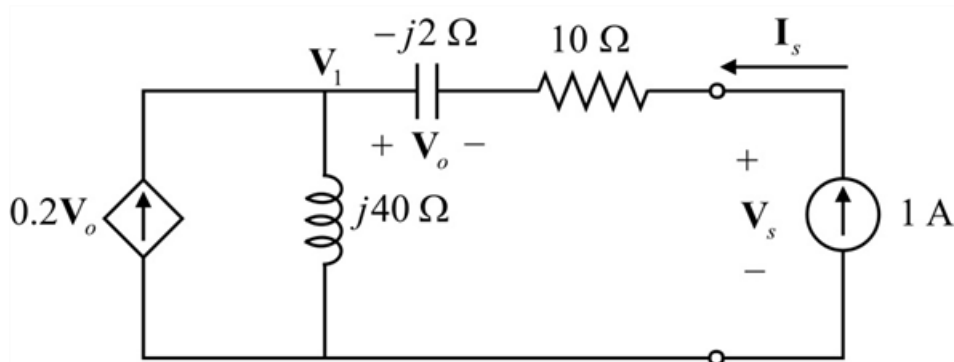


Figure 2

Step 3 of 5

Calculate the voltage $\mathbf{V}_o$ from the circuit in Figure 2.

$$\begin{aligned}\mathbf{V}_o &= -\mathbf{I}_s(-j2) \\ &= (-1)(-j2) \quad (\text{since } \mathbf{I}_s = 1 \text{ A}) \\ &= j2\end{aligned}$$

Apply Kirchhoff's current law at the node $\mathbf{V}_1$ for the circuit in Figure 2.

$$\begin{aligned}0.2\mathbf{V}_o &= \frac{\mathbf{V}_1}{j40} - \mathbf{I}_s \\ 0.2(j2) &= \frac{\mathbf{V}_1}{j40} - 1 \quad \left(\text{since } \begin{array}{l} \mathbf{I}_s = 1 \text{ A} \\ \mathbf{V}_o = j2 \text{ V} \end{array} \right) \\ \frac{\mathbf{V}_1}{j40} &= 0.4j + 1 \\ \mathbf{V}_1 &= j40(0.4j + 1) \\ &= -16 + j40\end{aligned}$$

Step 4 of 5

Apply Kirchhoff's voltage law for the circuit in Figure 2.

$$\begin{aligned}\mathbf{V}_1 - \mathbf{V}_o + 10\mathbf{I}_s - \mathbf{V}_s &= 0 \\ \mathbf{V}_s &= \mathbf{V}_1 - \mathbf{V}_o + 10\mathbf{I}_s\end{aligned}$$

Substitute $-16 + j40$ for $\mathbf{V}_1$, $j2$ for $\mathbf{V}_o$ and 1 A for $\mathbf{I}_s$ in $\mathbf{V}_s = \mathbf{V}_1 - \mathbf{V}_o + 10\mathbf{I}_s$

$$\begin{aligned}\mathbf{V}_s &= -16 + j40 - j2 + 10(1) \\ &= -6 + j38\end{aligned}$$

Step 5 of 5

Calculate the output impedance $\mathbf{Z}_o$.

$$\mathbf{Z}_o = \frac{\mathbf{V}_s}{\mathbf{I}_s}$$

Substitute $-6 + j38$ for $\mathbf{V}_s$ and 1 A for $\mathbf{I}_s$ in $\mathbf{Z}_o$.

$$\begin{aligned}\mathbf{Z}_o &= \frac{-6 + j38}{1} \\ &= -6 + j38 \Omega\end{aligned}$$

Therefore, the output impedance $\mathbf{Z}_o$ is $\boxed{-6 + j38 \Omega}$.